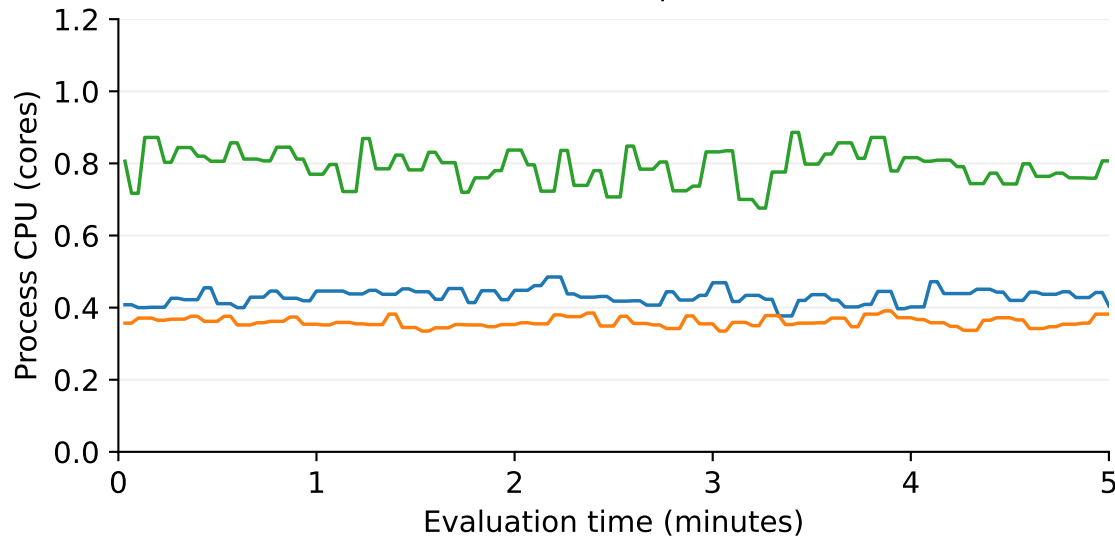


Consumer CPU usage

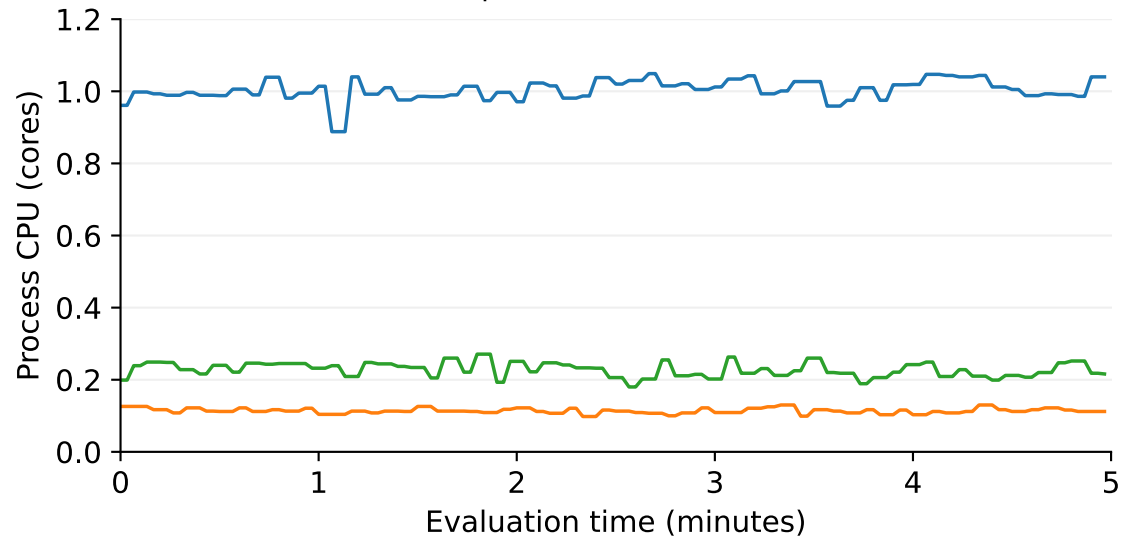
700 messages/s total · Three consumers · 80% of input across 12 of 60 partitions

Consumer 0 Consumer 1 Consumer 2

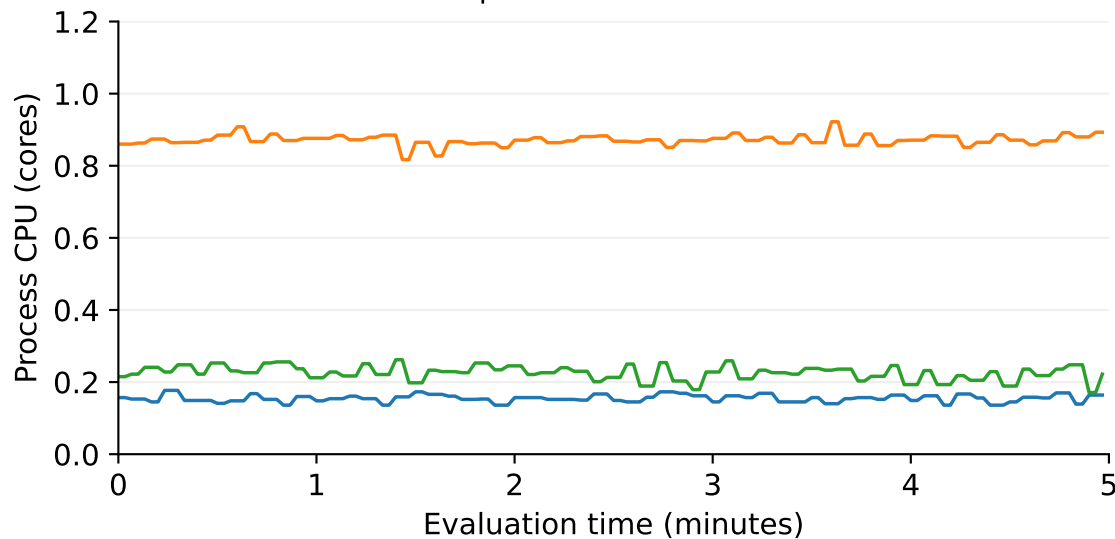
Distributed hot partitions



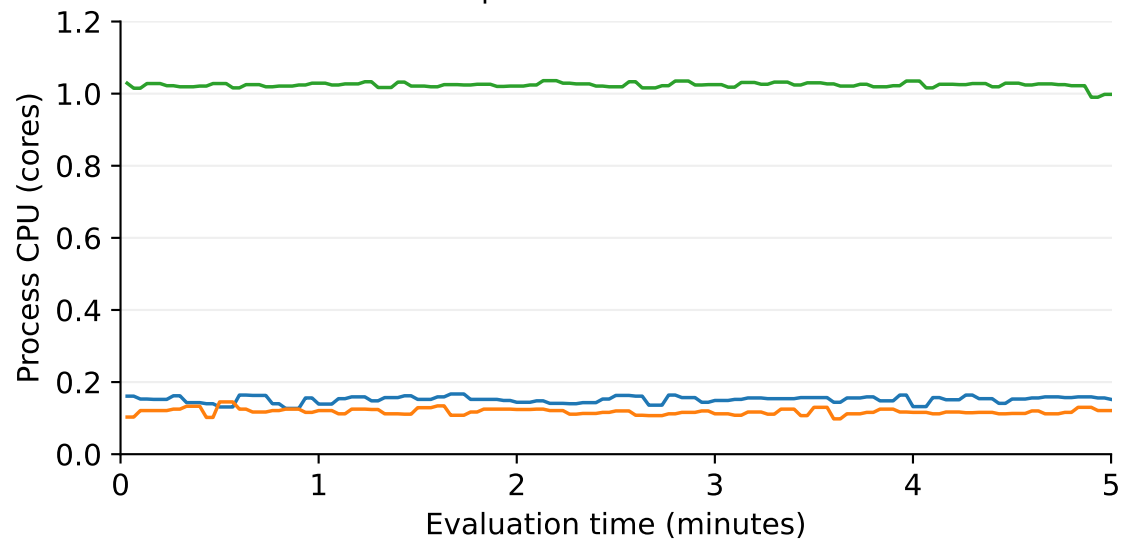
All hot partitions on Consumer 0



All hot partitions on Consumer 1



All hot partitions on Consumer 2



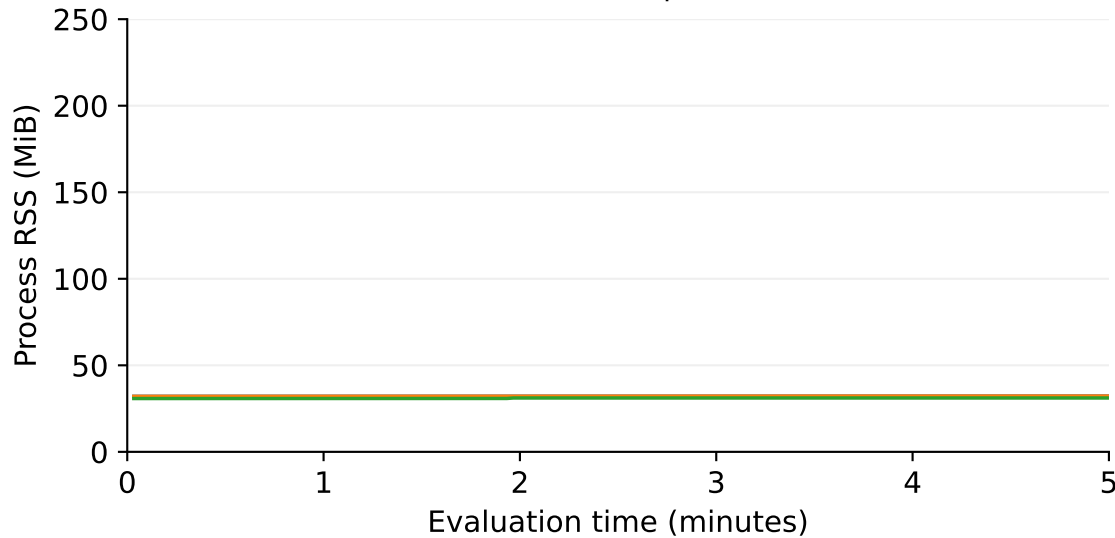
Five-minute evaluation. Main panels share the same vertical scale.
Process measurements; 1 CPU core corresponds to 100% CPU use.

Consumer memory usage

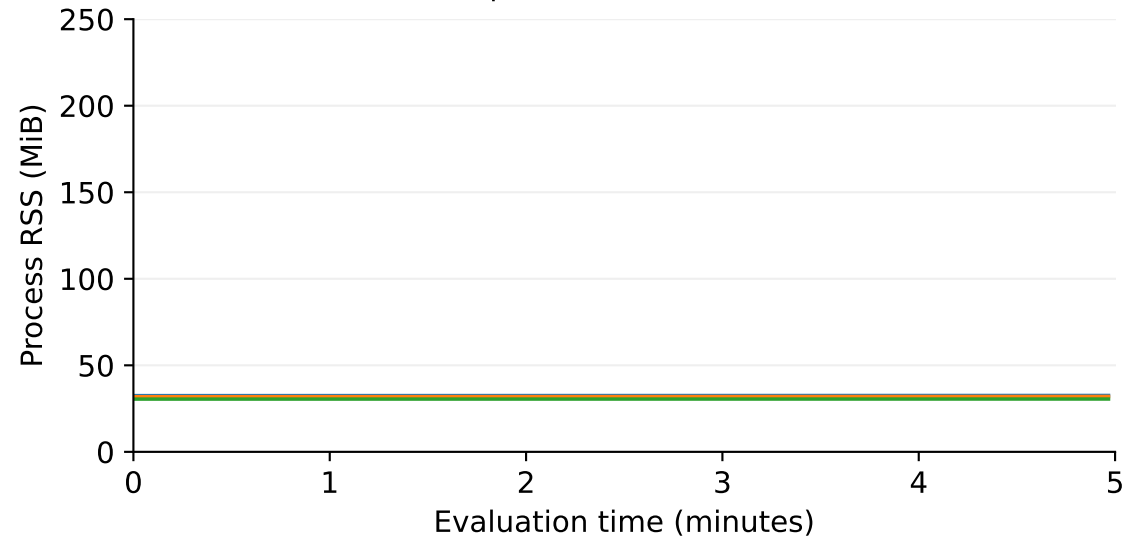
700 messages/s total · Three consumers · 80% of input across 12 of 60 partitions

Consumer 0 Consumer 1 Consumer 2

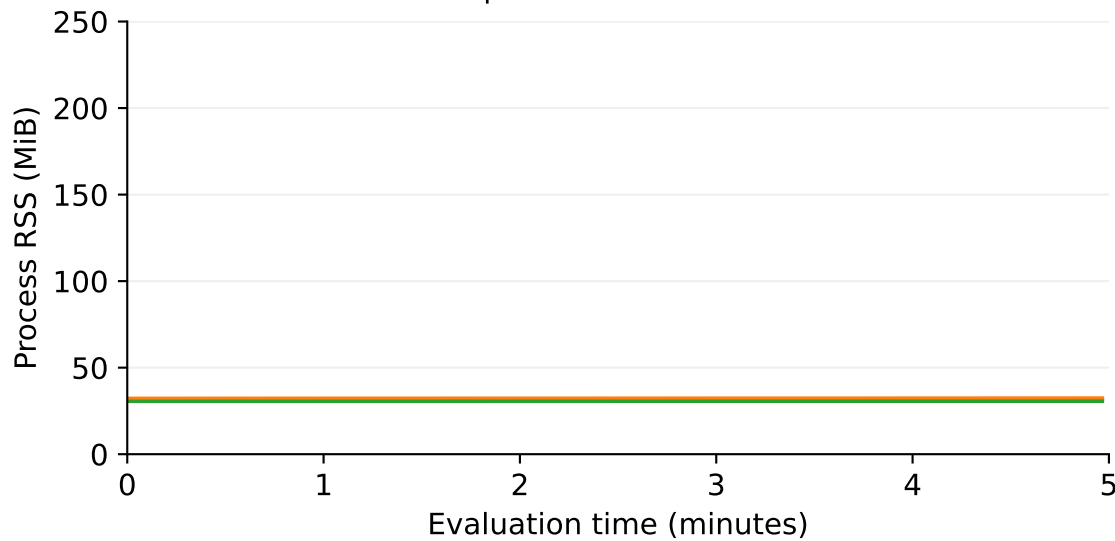
Distributed hot partitions



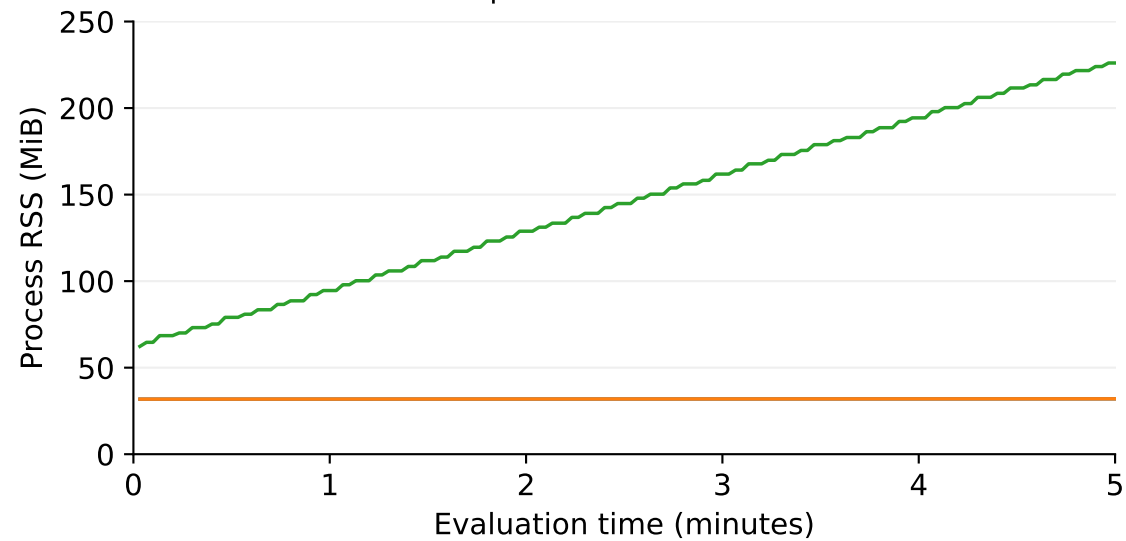
All hot partitions on Consumer 0



All hot partitions on Consumer 1



All hot partitions on Consumer 2



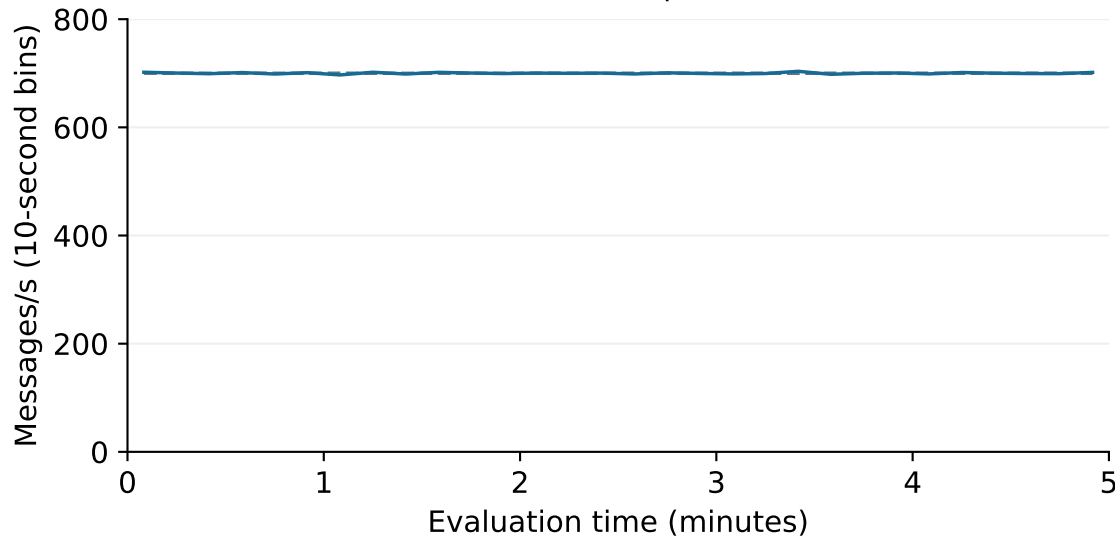
Five-minute evaluation. Main panels share the same vertical scale.
Memory is process resident set size (RSS).

Input and completion throughput

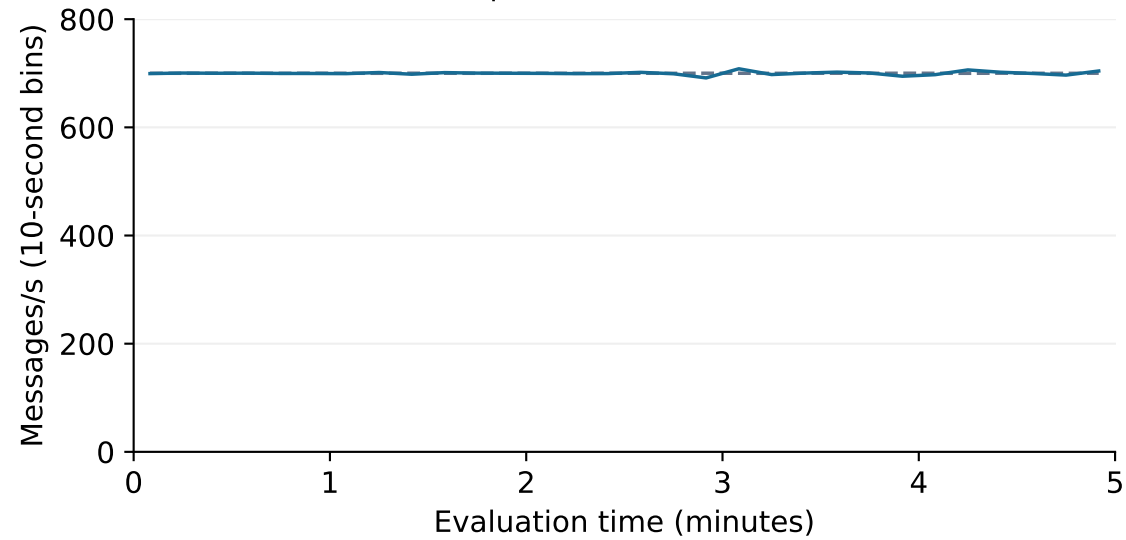
700 messages/s total · Three consumers · 80% of input across 12 of 60 partitions

--- Acknowledged input — Unique completions

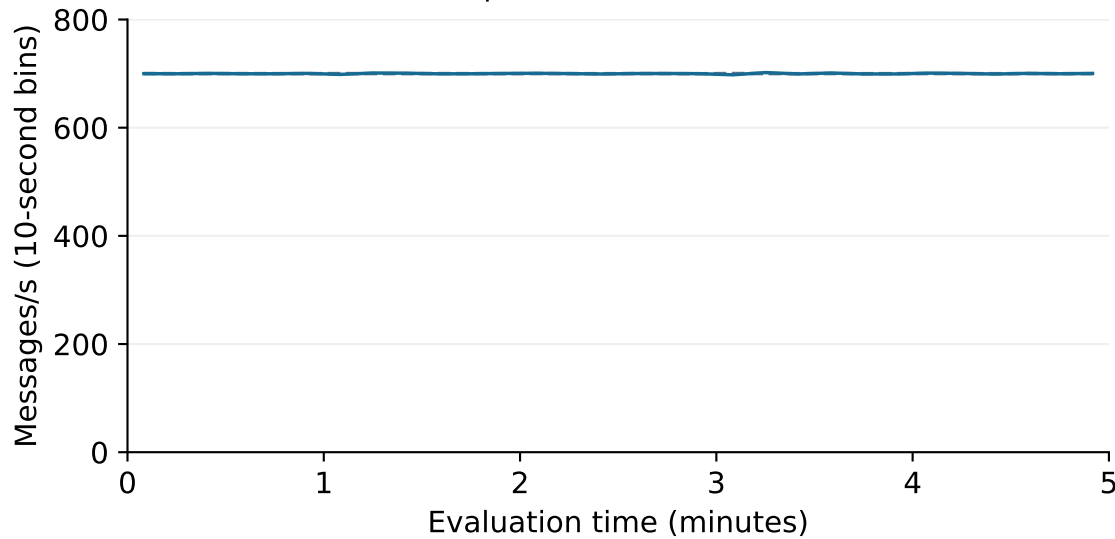
Distributed hot partitions



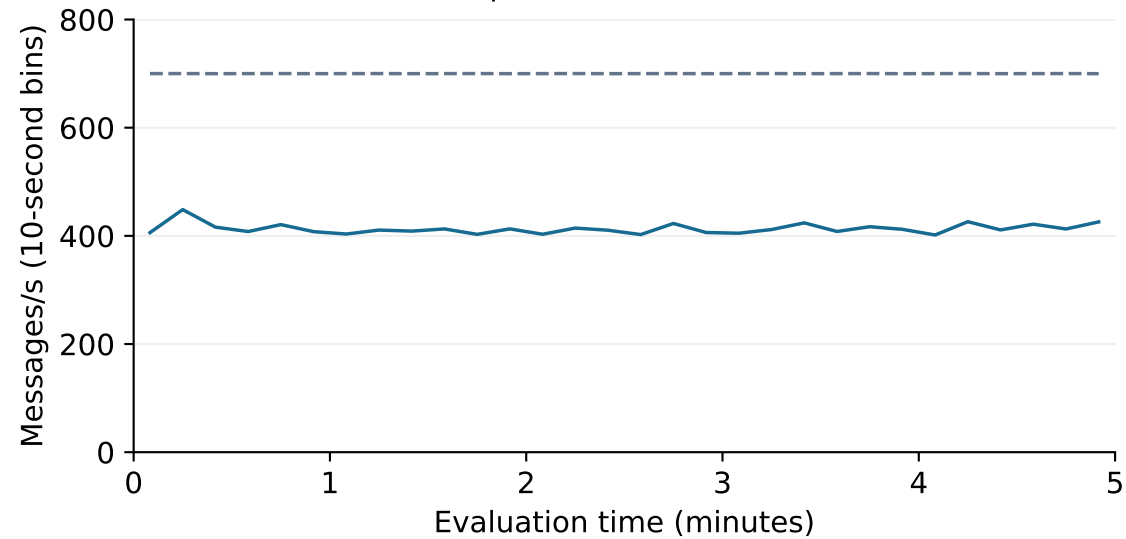
All hot partitions on Consumer 0



All hot partitions on Consumer 1



All hot partitions on Consumer 2

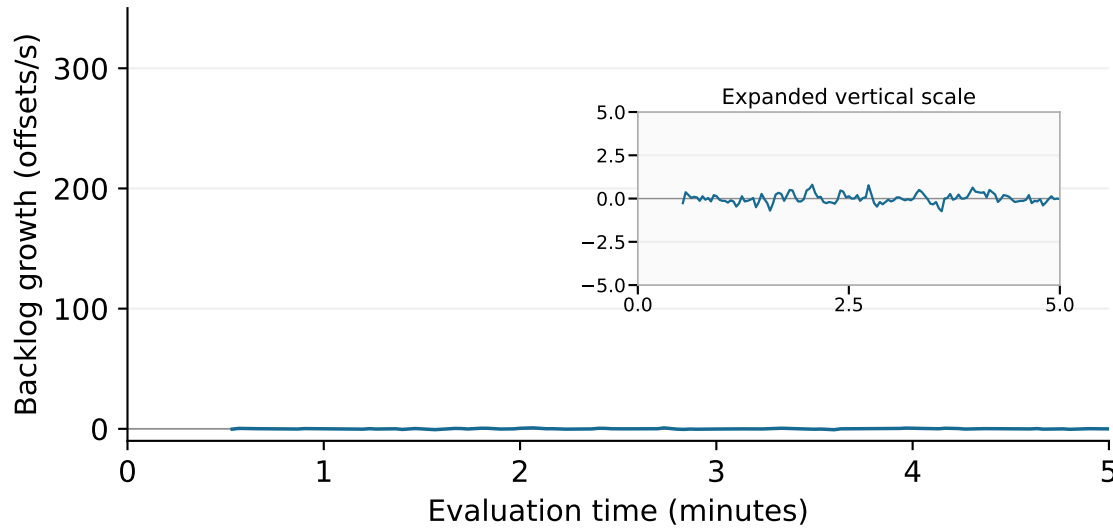


Five-minute evaluation. Main panels share the same vertical scale.
Unique completions can include warm-up records finishing during evaluation.

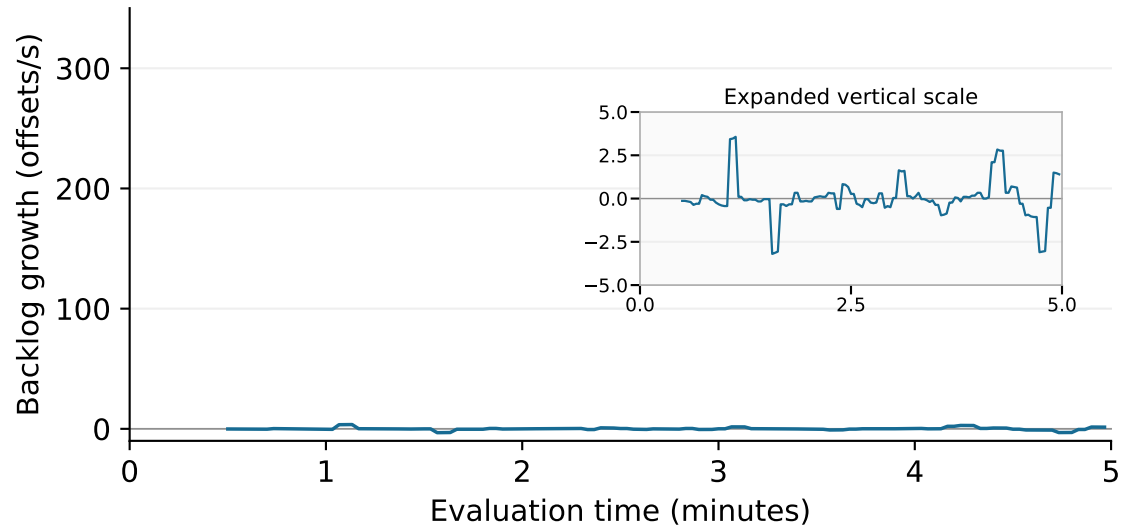
Processing-backlog growth

700 messages/s total · Three consumers · 80% of input across 12 of 60 partitions

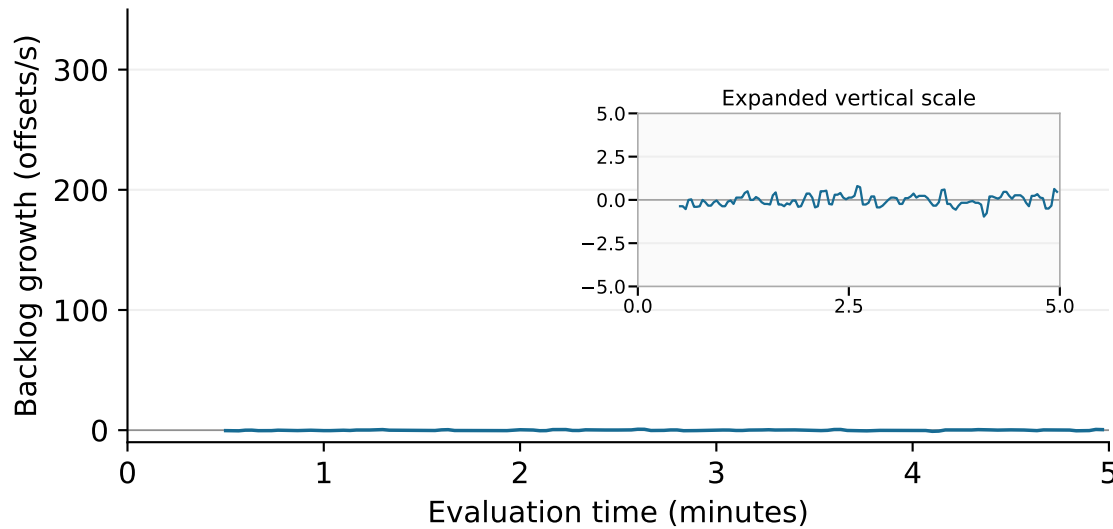
Distributed hot partitions



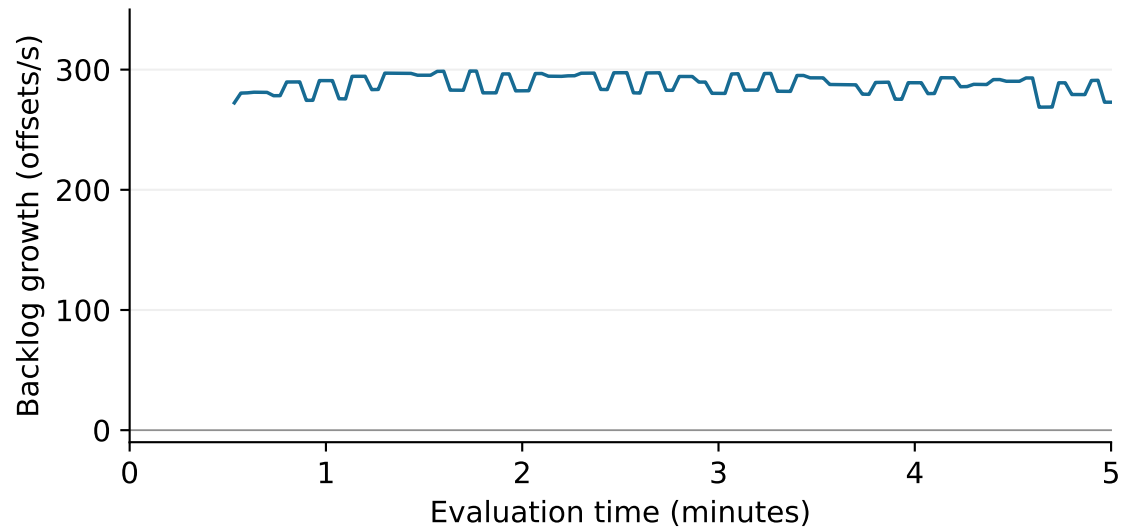
All hot partitions on Consumer 0



All hot partitions on Consumer 1



All hot partitions on Consumer 2



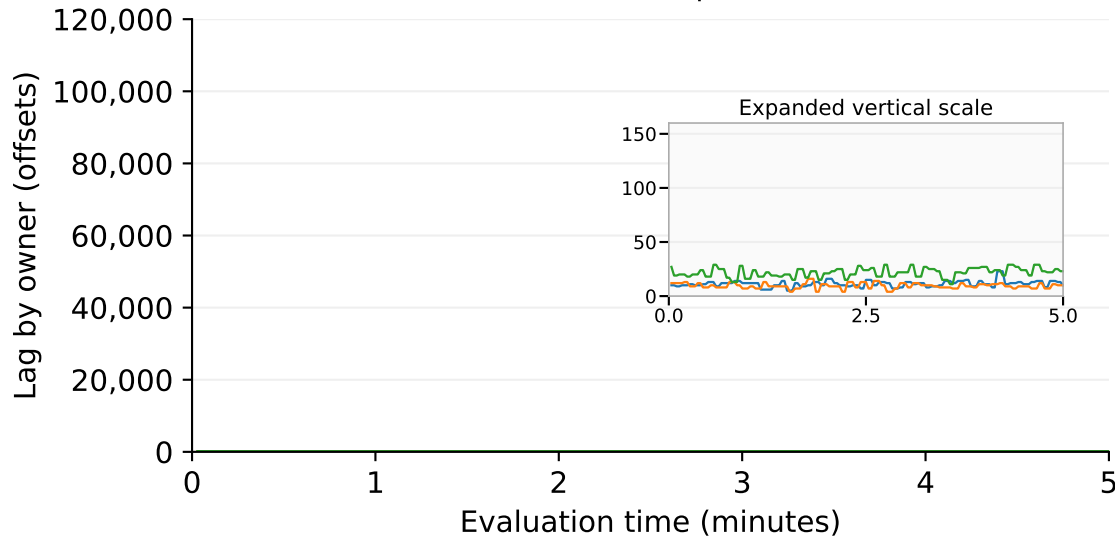
Five-minute evaluation. Main panels share the same vertical scale. Insets show the same data with a shared expanded scale. Growth uses a rolling 30-second window; gaps remain unavailable.

Lag by consumer

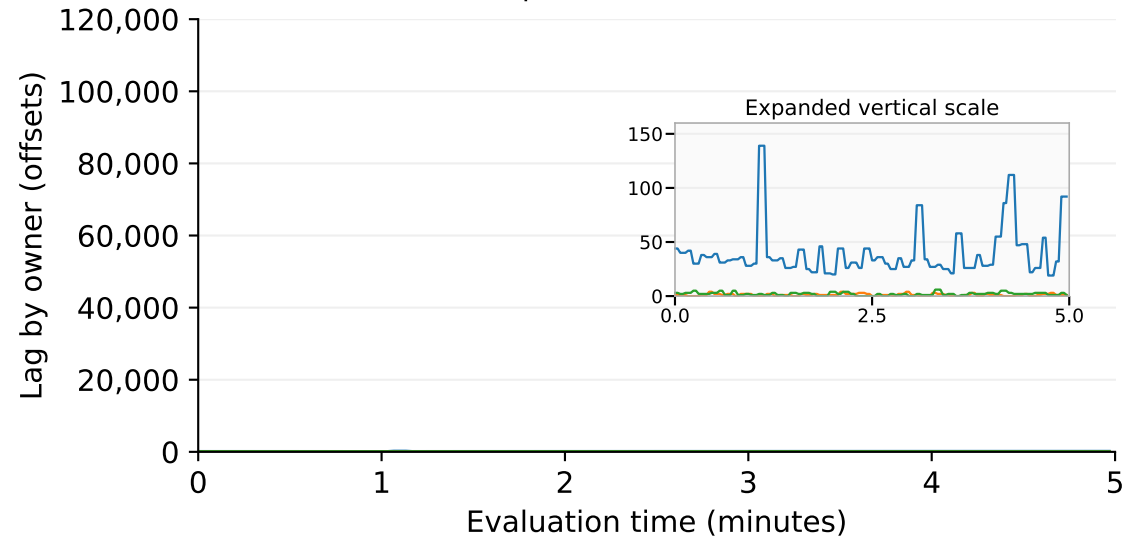
700 messages/s total · Three consumers · 80% of input across 12 of 60 partitions

Consumer 0 Consumer 1 Consumer 2

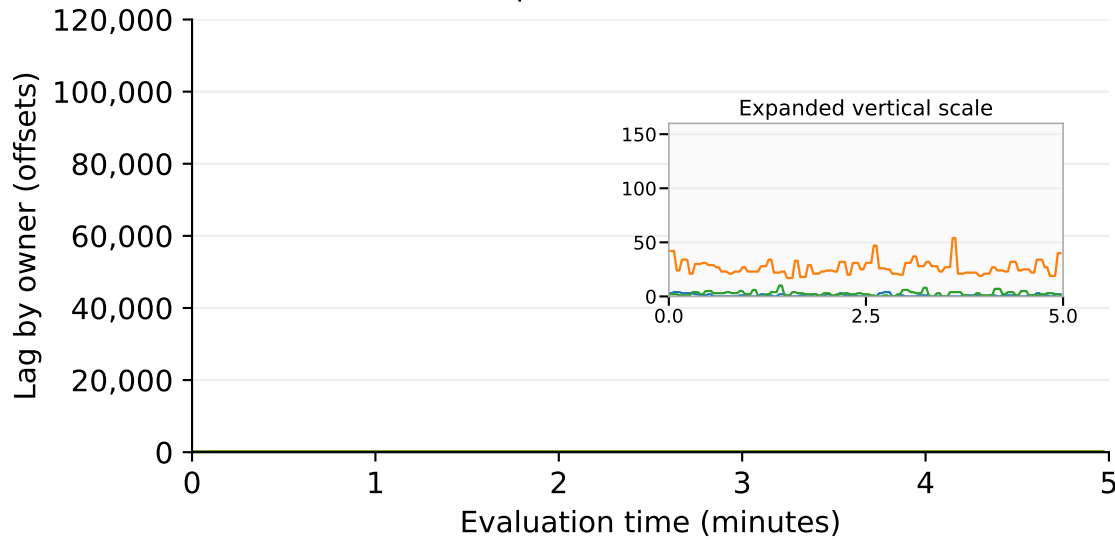
Distributed hot partitions



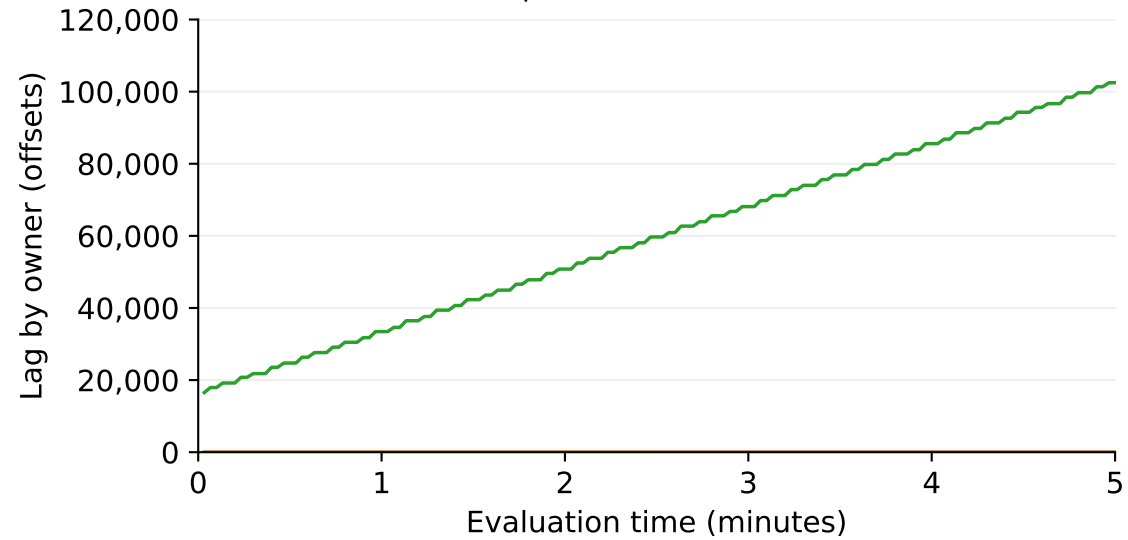
All hot partitions on Consumer 0



All hot partitions on Consumer 1



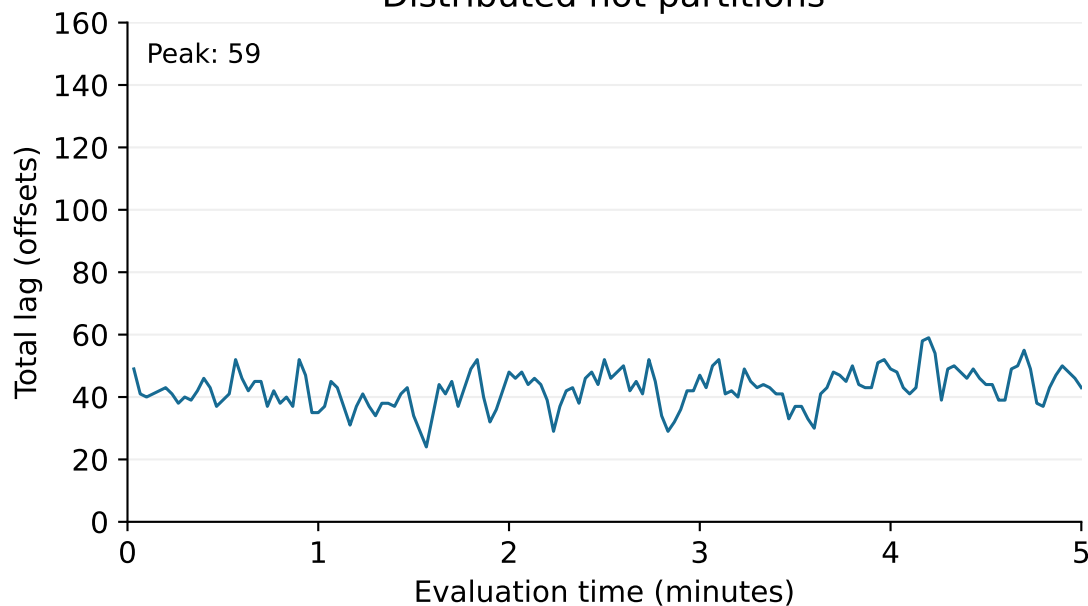
All hot partitions on Consumer 2



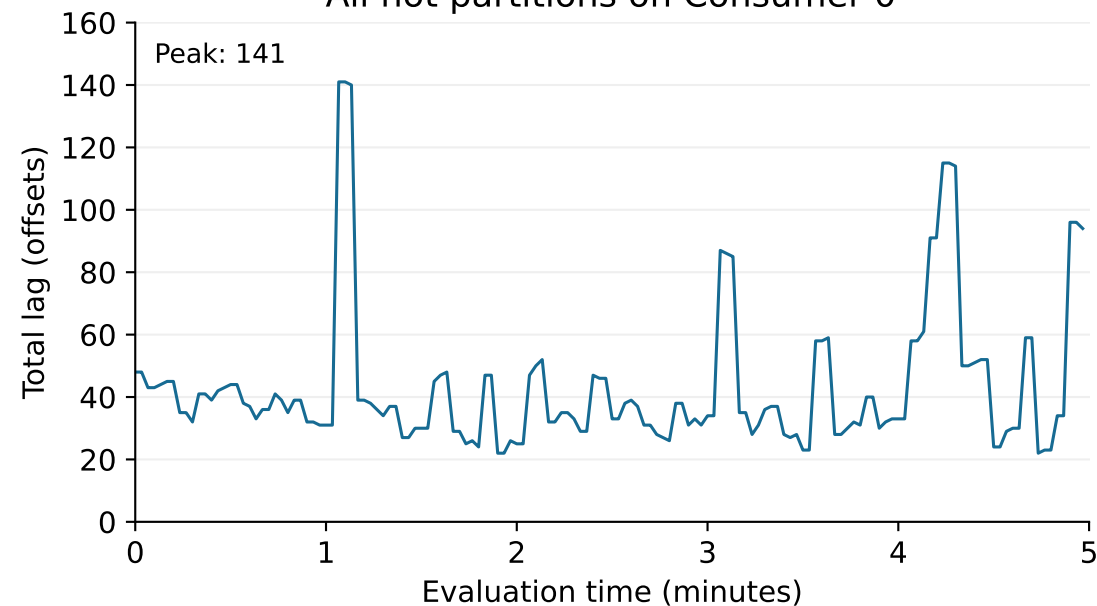
Five-minute evaluation. Main panels share the same vertical scale. Insets show the same data with a shared expanded scale.

Same 80/20 input, different ownership — 700 messages/s

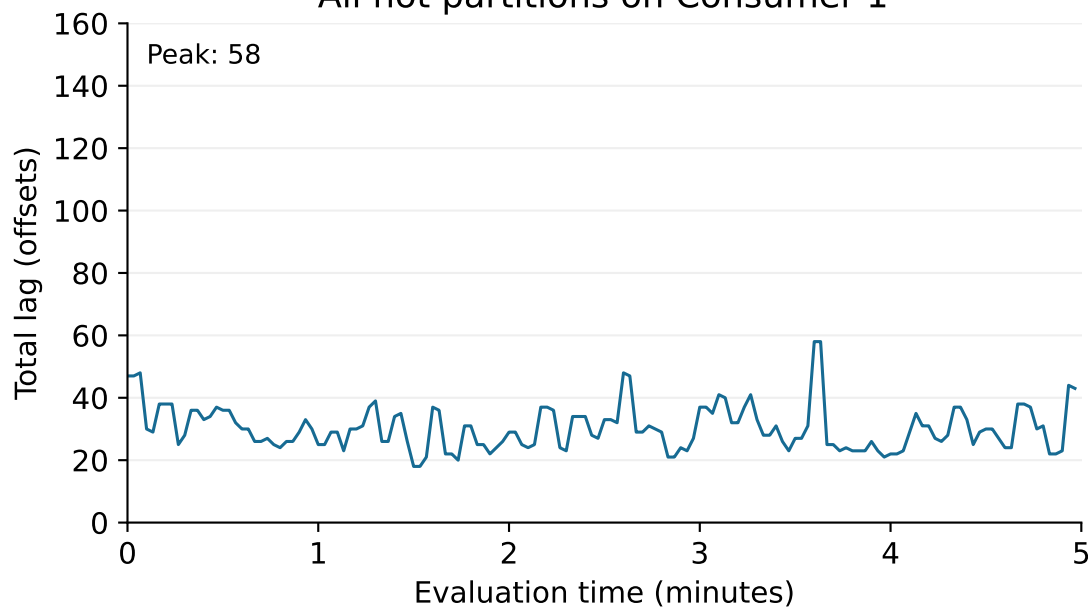
Distributed hot partitions



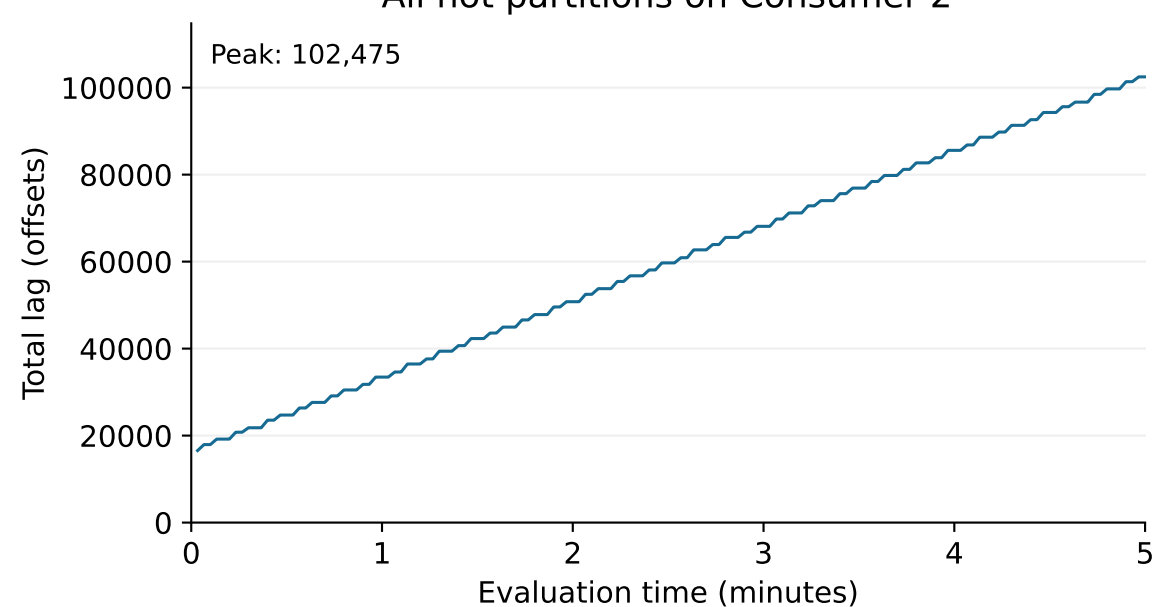
All hot partitions on Consumer 0



All hot partitions on Consumer 1

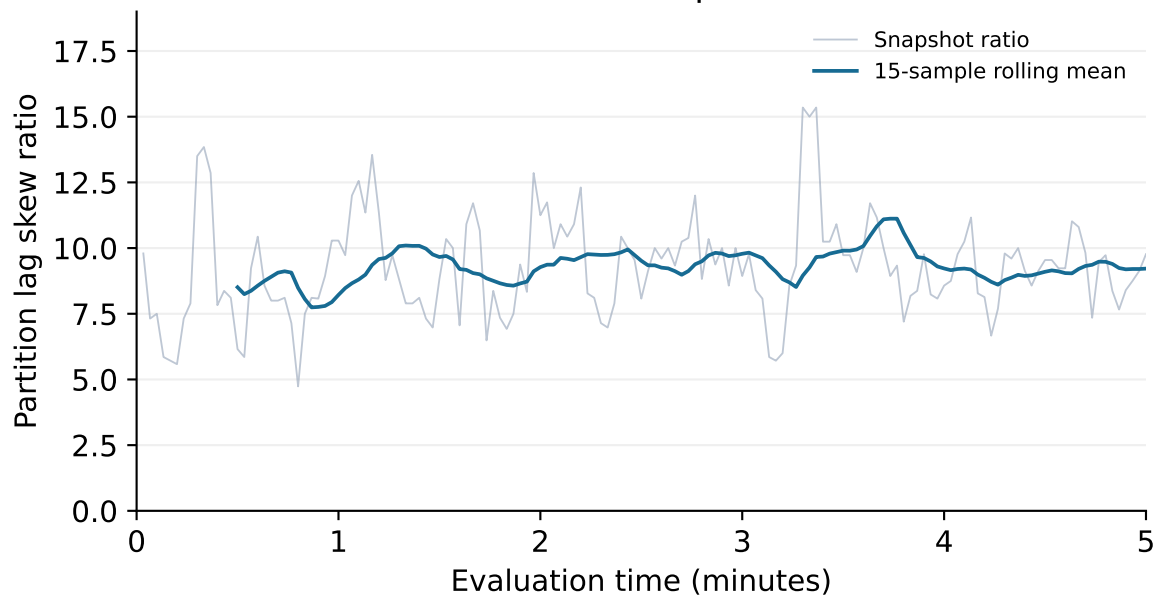


All hot partitions on Consumer 2

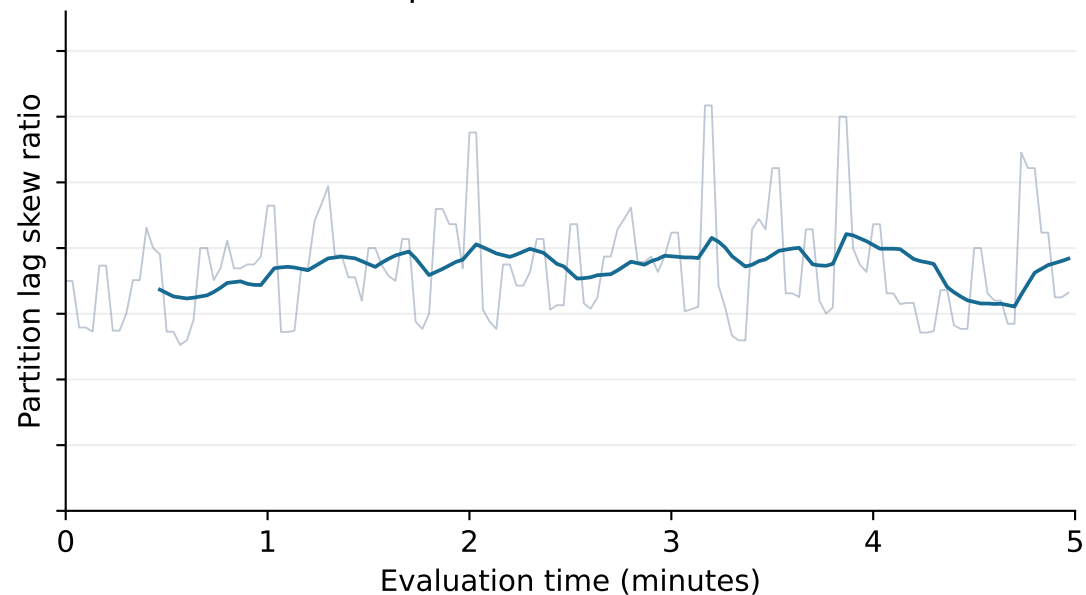


Partition lag skew — 700 messages/s, three consumers

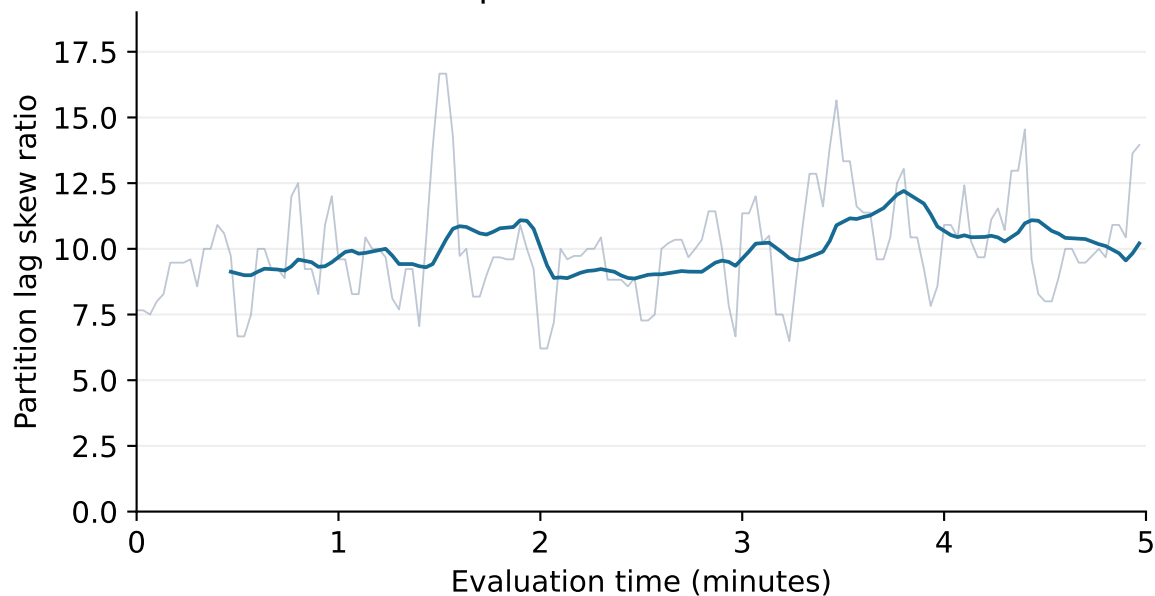
Distributed hot partitions



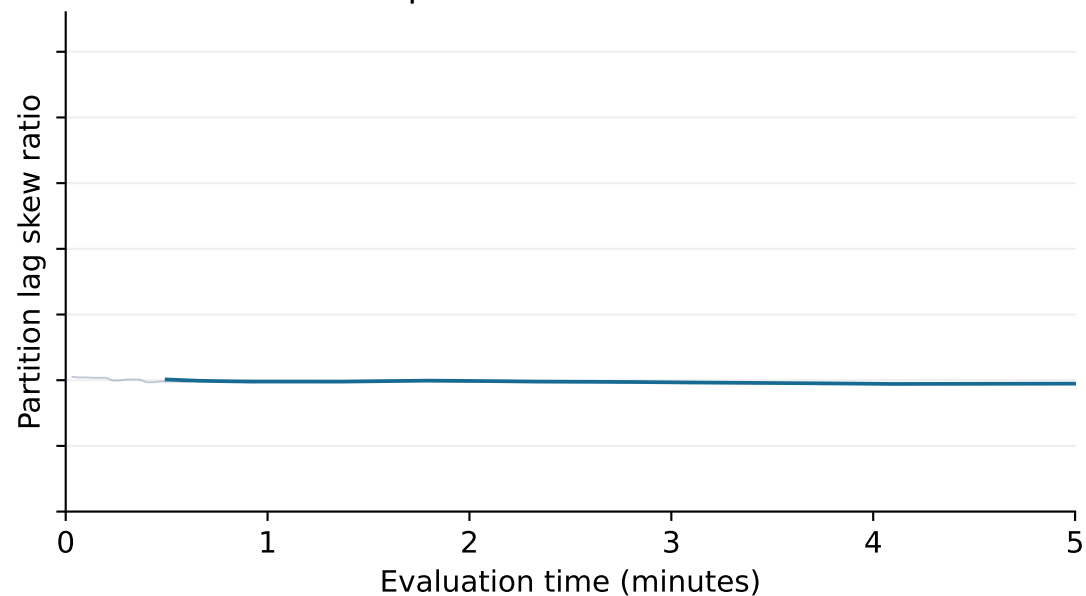
All hot partitions on Consumer 0



All hot partitions on Consumer 1



All hot partitions on Consumer 2



Ratio = maximum partition lag / mean partition lag; zero if all lag is zero. Rolling mean: 15 consecutive valid samples.